

Transcription: Study Notes

Session Objectives:

SOM.MK.I.BPM1.1.FTM.3.BCHM.0049	Describe the structural features of a gene and their functions
SOM.MK.I.BPM1.1.FTM.3.BCHM.0050	Describe the types and functions of the different cellular RNAs (mRNA, tRNA, rRNA, snRNA, miRNA)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0051	Describe the major steps of RNA synthesis in prokaryotes and eukaryotes
SOM.MK.I.BPM1.1.FTM.3.BCHM.0052	Explain the function and structure of DNA sequences that are involved in the initiation (promoters) and termination of transcription
SOM.MK.I.BPM1.1.FTM.3.BCHM.0053	Predict the sequence of the transcribed RNA if the template DNA strand or the coding strand of the RNA is provided
SOM.MK.I.BPM1.1.FTM.3.BCHM.0054	List the RNA products that are transcribed by the three different types of RNA polymerases in eukaryotes
SOM.MK.I.BPM1.1.FTM.3.BCHM.0057	Explain the mechanism of action of rifampin and alpha-amanitin as inhibitors of transcription.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0058	Describe the effect of the antibiotic drugs that interfere with DNA gyrase such as (quinolones) ciprofloxacin

Assigned Readings: Lippincott's Illustrated Reviews Biochemistry. Chapter 31. [RNA Structure, Synthesis, and Processing | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library \(lwwhealthlibrary.com\)](#)
Questions 31.1 – 31.5. Concept Map on Page 494.

IMPORTANT: Panopto videos are available on the topic of Transcription.

Transcription is the process of the conversion of DNA encoded information to its RNA coded equivalent. Transcription involves the polymerization of ribonucleosides guided by complementary base pairing to produce an RNA transcript of the gene. The template used for RNA synthesis is only one strand of the DNA double helix that composes the gene.

The DNA strand serving as the **template for transcription** is called the **antisense strand** and the **other strand** is called the **sense strand**. The sense strand of the DNA will have the same sequences as the resulting RNA with the exception that uracil replaces thymine.

There are four different types of RNA molecules:

1. **mRNA (messenger RNA)** encodes the amino acid sequence of a polypeptide. mRNAs are the transcripts of protein coding genes.
2. **tRNA (transfer RNA)** brings amino acids to the ribosome during the translation process.
3. **rRNA (ribosomal RNA)** combines with ribosomal proteins to form the ribosome. mRNA is translated into protein by the ribosome.
4. **snRNA (small nuclear RNA)** combines with certain proteins and is involved in RNA processing (example, SNRNPs and mRNA splicing) and gene regulation in eukaryotes (eg. microRNAs).

The enzyme responsible for transcription is called **RNA polymerase**.

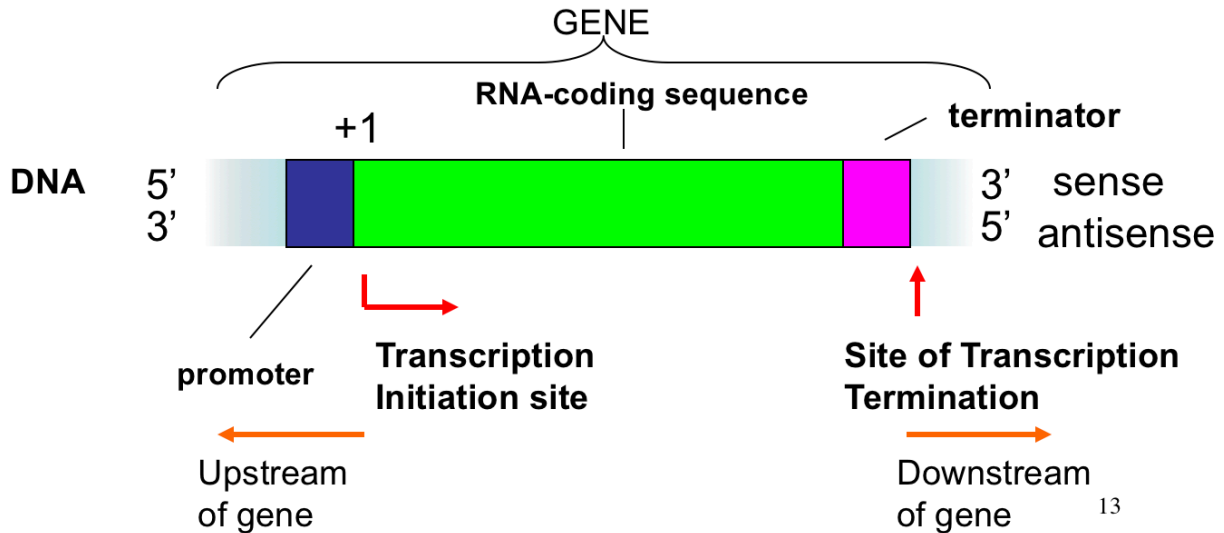
In bacteria, there is only a single RNA polymerase that is composed of a core polymerase protein plus another protein called ***sigma factor***.

In eukaryotes there are three different RNA polymerases each of which transcribes a different type of RNA:

1. **RNA polymerase I**: 5.8S, 18S, and 28S rRNA genes
2. **RNA polymerase II**: all protein-coding genes (mRNA) and some snRNAs
3. **RNA polymerase III**: tRNA genes and some snRNAs

Anatomy of a Gene:

A gene is comprised of three main elements: promoter, RNA coding sequence and termination sequence.



Regions of a prokaryotic gene:

1. **Promoter:** located upstream of the RNA coding sequence, and ensures the proper location of transcription initiation.
2. **RNA coding sequence:** the DNA sequence that is transcribed into RNA.
3. **Terminator:** a sequence downstream of the RNA coding sequence and specifies where transcription will stop.

The initiation of transcription involves the binding of the RNA polymerase and various transcription factors to DNA sequences located in the **promoter** of the gene. The promoter is located 5' (or upstream) to the region of the gene encoding the protein.

The prokaryote promoter sequences are generally found at -35 and -10 bp (upstream) from the transcription start site: -35 consensus sequence: 5'-TTGACA-3' and the -10 consensus sequence: 5'-TATAAT-3' (also called the **Pribnow box**)

Eukaryote promoters typically consist of a group of short conserved sequences. Some of these are the TATA box (5'-TATAAA-3'), GC box (5'-GGGCGG-3') and the CAAT box (5'-CCAAT-3').

Initiation of Transcription:

Transcription initiation in prokaryotes involves the following steps:

- 1) Binding of RNA polymerase and sigma factor to -35 and -10 regions of promoter.
- 2) Unwinding of the DNA to expose regions of single stranded DNA.
- 3) Polymerization of ribonucleosides to form a RNA transcript using the antisense template.
- 4) Dissociation of sigma factor from RNA polymerase (after 8-9 bp are synthesized).
- 5) Polymerization proceeds as RNA polymerase unwinds DNA.

- 6) Transcription is terminated upon encountering a termination sequence.

Transcription in eukaryotes involves an RNA polymerase and a large number of different proteins called transcription factors (a few mentioned below).

- 1) TATA binding protein (TBP) and Transcription Factor IID (TFIID) bind to promoter at the TATA box.
- 2) A number of other different transcription factors along with RNA polymerase bind to the promoter and the transcription initiation site.
- 3) One of these transcription factors, TFIIF, unwinds DNA via helicase activity.
- 4) Additionally, TFIIF activates the RNA polymerase by phosphorylation and transcription is initiated.
- 5) Most transcription factors are released following initiation of transcription.

The movement of the RNA polymerase along the DNA during transcription results in supercoiling of the DNA: +ve supercoils in the front of the RNA polymerase and -ve supercoils behind the RNA polymerase. **These supercoils are corrected by the enzymes gyrase and topoisomerase** in prokaryotes, respectively. In eukaryotes these enzymes are called Topoisomerase I and Topoisomerase II.

Termination of Transcription:

The termination of transcription can involve both **intrinsic** factors and **extrinsic** factors.

Intrinsic Termination: Located at the 3' end of the transcribed RNA is a region containing palindromic sequences. These sequences contain regions that are complementary to each other and as a result they can anneal and form a secondary stem loop structure. The stem loop structure includes a G-C rich region followed by several ribonucleotides of uracil. This stem loop structure, also called the termination structure, is recognized by the RNA polymerase and causes the RNA polymerase to dissociate from the DNA and the transcribed RNA.

Extrinsic Termination: Extrinsic termination involves the binding of a protein called the **Rho protein** to the newly transcribed RNA molecule. The Rho protein attaches to the RNA molecule and moves along the RNA following the RNA polymerase. The RNA polymerase will pause once it synthesizes the region of the RNA containing the termination sequences (those that produce the stem loop structure). Once the Rho protein reaches the termination structure it unwinds the DNA-RNA hybrid and pulls the RNA from the RNA polymerase. This causes the release of RNA polymerase, Rho protein and the RNA.

Clinical Relevance:

Transcription of RNA is a fundamental process that is necessary for life in all organisms. It is necessary for the production of mRNA that will ultimately be translated into protein. Thus, defects that affect the transcriptional machinery are typically lethal to the cell and the organism. As a result, clinical pathologies are typically not observed.

This makes the transcriptional machinery and the transcription process a suitable target if one wanted to compromise an organism. This strategy has been used in the development of antibiotic drugs to kill pathological bacteria.

Rifamycin is an antibiotic derived from *Streptomyces* (a fungi-like organism). It inhibits the initiation of transcription by binding to RNA polymerase and blocking the formation of the first phospho-diester bond of the RNA. The RNA polymerase of eukaryotes is not affected.

Actinomycin D is another antibiotic derived from *Streptomyces*. This antibiotic binds tightly to double stranded DNA and inhibits both DNA replication and transcription. It does this by preventing the formation of regions of single stranded DNA. It is effective in both prokaryotes and eukaryotes.

Bacterial gyrase (topoisomerase) is inhibited by antibiotics such as the coumarins (including novobiocin) and the quinolones (including nalidixic acid and ciprofloxacin).

An instance whereby transcription may be seriously disrupted in a human patient can occur when they consume the poisonous mushroom, *Amanita phalloides* (The Death Cap Mushroom). This mushroom contains the toxin alpha amanitin that is an inhibitor of RNA polymerase II. As a result, cells exposed to this toxin cannot produce mRNA. Initially, the effect is seen in the cells of the intestine, and later in the liver and kidney – leading to renal and hepatic failure and often death.

Poisoning by *Amanita phalloides* is complicated by the fact that this mushroom also contains high concentrations of the molecule phalloidin. Phalloidin is a potent actin (a cellular cytoskeletal molecule) binding agent and prevents microfilament depolymerization. This primarily affects stability of liver cells leading to liver failure.

DNA replication and transcription have a number of similar features as both processes involve the synthesis of nucleic acid polymers. There are however some important differences that you should be aware of that are shown in the table below:

Comparison of DNA Replication and Transcription:

RNA

RNA polymerase
NTPs precursor
no primer needed
uracil pairs with adenine

DNA

DNA polymerase
dNTP precursor
primer required for initiation
thymine pairs with adenine